

Zirui Wen

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Education

Stevens Institute of Technology, MS in Applied Artificial Intelligence, GPA 3.94/4.0	Sep 2024 – May 2026
University of Birmingham, BS in Applied Mathematics with Information Computing Science	Sep 2020 – Jun 2024
Jinan University, BS in Information and Computing Science	Sep 2020 – Jun 2024

Publications

Shihao Yang*, **Zirui Wen***, Wenxin Zhan*. et al., “Knowledge graph representation of the mappings between seizure semiology and epileptogenic zones,” Scientific Reports, 2026

Zirui Wen*, Junjie Zhang*, and Yuhao Zhang*, “Covid-19 infection prediction using physical signs,” in International Conference on Cloud Computing, Performance Computing, and Deep Learning (CCPCDL 2022), SPIE, vol. 12287, 2022, pp. 575–589

Manuscripts

Qingyang Yu*, Hao Wang, **Zirui Wen**, et al. A Chance-Constrained Optimization Framework for LLM Forgetting. Under review at ICML, 2026.

Zirui Wen*, Shihao Yang*, Feng Liu. et al. Uncovering Epileptic Seizure Propagation Using Knowledge Graph-based Reinforcement Learning. Under review at IJSE Transactions, 2026.

Zirui Wen*, Wensheng Gan. Automatic Prompt Optimization for Medicine, Outstanding Bachelor’s Thesis, 2024.

Research & Work Experience

Research Assistant Nov 2025 – Present

Intelligent System (IntelliSys) Lab, Stevens Institute of Technology, Supervised by Dr. Hao Wang

Paper: A Chance-Constrained Optimization Framework for LLM Forgetting (Under review at ICML)

- Proposed a **chance-constrained optimization** framework for LLM continual post-training, modeling forgetting as a probabilistic risk and mitigating it by up to **2.96%** on **MMLU** for **7B-14B** models.

Research Assistant Nov 2024 – Present

Brain Imaging and Graph Learning Lab, Stevens Institute of Technology, Supervised by Dr. Feng Liu

SEIZ EEG Labeling (Collaboration with Rutgers)

- Built a Transformer-based SEIZ EEG sequence labeling pipeline over 670 samples and 1.07M time points, achieving strong clinical-level seizure detection performance in a reproducible setting.

Clinical Epilepsy QA System (GraphRAG & LLM Fine-tuning)

- Designed a hybrid **GraphRAG** pipeline integrating **Neo4j Cypher-based structural** queries with **FAISS dense vector retrieval** to provide factual grounding and mitigate LLM hallucinations in the medical domain.
- Fine-tuned **LLaMA-3** and **Mistral-7B** models via **QLoRA** on custom domain-specific instruction datasets, achieving a **20%** improvement in factual consistency on internal benchmarks over CoT baselines.

Paper: Knowledge graph representation of the mappings between seizure semiology and epileptogenic zones (Published at Scientific Reports)

- Engineered a **GPT-4o** extraction pipeline with strict constraints to parse **10k+ unstructured clinical records** into 172 standardized entity nodes and 69 region nodes.
- Constructed a probabilistic Knowledge Graph by formulating a **Poisson-Gamma Bayesian smoothing** model to regularize sparse co-occurrences and applying **Gaussian Mixture Models (GMM)** to cluster latent relationship strengths.

Paper: Uncovering Epileptic Seizure Propagation Using Knowledge Graph-based Reinforcement Learning (Under review at IJSE Transactions)

- Modeled seizure progression as a discrete-state decision **MDP** process over semiology transitions and designed a reward that prefers early and stable predictions under partial observations.
- Trained with a **PPO-style** objective and achieved a **40 %** improvement in predictive return with improved cross-patient generalization compared to baseline MC methods.

Data Team Assistant Intern Nov 2023 – Mar 2024

Siemens

- Built modular **Python**, **SQL** and **Excel** pipelines for multi-source data ingestion, cleansing, validation, and KPI analysis. Automated data processing workflows, substantially reducing manual effort while improving reporting consistency and

data quality.

Research Assistant

Mar 2024 – May 2024

Pazhou Lab, Jinan University, Supervised by Dr. Wensheng Gan

Paper: Automatic Prompt Optimization for Medicine, Outstanding Bachelor's Thesis, 2024

- Investigated **gradient-based prompt optimization** for large medical language models, formulating prompt tuning as a continuous optimization problem over text embeddings to improve factual grounding and reasoning consistency.
- Proposed a **text-gradient update mechanism** that propagates feedback from model factuality and faithfulness scores back into the prompt representation, enabling end-to-end optimization without manual intervention.
- Incorporated a Bayesian reliability estimator to assess and reweight candidate prompts by posterior confidence, yielding 12% improvement in factual precision compared with **COT** and a noticeable reduction in hallucination rate on internal medical QA benchmarks.

Projects

Miniagent (Personal Project) Jan 2026

- Built a modular **RL(GRPO) agent training framework** for tool-augmented tasks, providing a clean training skeleton plus a **tool-call** interface to plug in external **web search** tool during rollouts for end-to-end agent RL experiments.

Mini-vLLM (Personal Project) Dec 2025

- Implemented a lightweight **vLLM-style** inference engine with **PagedAttention** and block-based KV cache management to reduce memory fragmentation and improve long-context decoding throughput.

Smarter Doctor agent (Google Cloud Hackathon) Oct 2025

- Built and deployed a real-time voice-based **medical agent** on **Google Cloud** using a FastAPI backend and Next.js frontend with WebSocket streaming and tool orchestration, integrating Twilio, **Elasticsearch**, **BigQuery**, and **Gemini** through **Vertex AI**, with CI/CD via Docker and GitHub Actions.

Kernel K-Means GPU Accelerator (Class Project) May 2025

- Investigated computational bottlenecks in **kernel K-Means** clustering, identifying the $O(n^2)$ kernel matrix construction and dense updates as the primary scalability constraints.
- Implemented a **sparse linear algebra** reformulation of the kernel update step, expressing centroid computations as batched sparse–dense matrix multiplications (SpMM) to exploit **GPU parallelism**.
- Designed a hybrid memory layout using **blockwise CSR storage** and **shared-memory caching**, reducing kernel recomputation and achieving up to $8.3\times$ speedup over a CUDA baseline on 100k-sample datasets.

Liar's Bar: Game Theory (Class Project) Dec 2024

- Formulated the bluffing game as an extensive-form, **imperfect-information game**, constructing a payoff and utility model to mathematically represent truthful play, bluffing, and calling strategies from the game rules.
- Derived the mixed-strategy **Nash equilibrium** and **subgame perfect equilibrium** (SPNE) using backward induction over the full game tree, achieving closed-form optimal bluffing probabilities.
- Incorporated **Bayesian belief** updating into player decision making to dynamically adjust bluffing probability based on observed actions; **Monte-Carlo simulations** demonstrated higher win rate over non-Bayesian strategies.

COVID-19 Infection Prediction Using ML (Research Project) Nov 2022

Paper: COVID-19 Infection Prediction using Physical Signs (Published at CCPCDL 2022)

- Vitals-based COVID-19 classification with **Decision Tree** / **Random Forest** / **SVM** under K-fold CV: Random Forest achieved 99.8% accuracy and $AUC = 0.98$, with oxygen saturation identified as the dominant feature by permutation importance.

Asset Allocation Optimization Based on PSO and fixed point method (Class Project) Jun 2022

- Framed portfolio allocation in the **Markowitz mean–variance** setting with practical constraints (target return, weights sum-to-one, no short-selling) and evaluated performance with a sliding-window backtest and cumulative-return curves.
- Proposed a hybrid solver: use **PSO** for global exploration, then convert constraints via an indicator-function formulation and solve with a **fixed-point proximal** (FPPA) update; additionally reformulated a sparse **SSMP variant with L1 regularization** and more stable convergence and stronger cumulative-return $5\times$ than baseline PSO across windows.

Research Interests & Technologies

Research Interests: LLM Reasoning and Planning, Reinforcement Learning, World Models, Agents, Multimodal, Continual Learning, Federal Learning, Knowledge Graph Reasoning, Probabilistic Modeling, ML Systems, AI4health, Embodied AI

Technical Expertise: Post-training and alignment of LLMs; Reinforcement learning for reasoning and decision-making; Language agents with tools, memory, and retrieval; Personalized Federal Learning; Continual learning and forgetting mitigation; Transformer-based sequence modeling; Efficient inference and GPU-accelerated ML systems

Programming & Tools: Python, C++, SQL, R, CUDA; PyTorch, TensorFlow, scikit-learn, Hugging Face, vLLM; Neo4j, FAISS, Elasticsearch; Linux, Docker, Git, MLflow, Optuna, PostgreSQL, MySQL, FastAPI. GCP, AWS